

Foundation (Jalapeno)

- Q1.** A basketball team scored 48 points. Is 48 divisible by 3?
(A) Yes
(B) No
(C) Maybe
(D) Depends on the opponent
(E) Not enough information
- Q2.** The distance between two tennis courts is 150 meters. Is 150 divisible by 5?
(A) Yes
(B) No
(C) Maybe
(D) Depends on the court size
(E) Not enough information
- Q3.** A cyclist traveled 63 kilometers. Is 63 divisible by 9?
(A) Yes
(B) No
(C) Maybe
(D) Depends on the route
(E) Not enough information
- Q4.** A marathon runner completed 120 miles. Is 120 divisible by 10?
(A) Yes
(B) No
(C) Maybe
(D) Depends on the runner's pace
(E) Not enough information
- Q5.** A gymnast performed 18 routines. Is 18 divisible by 2?
(A) Yes
(B) No
(C) Maybe
(D) Depends on the routine type
(E) Not enough information
- Q6.** The score in a football game is 75 to 42. Is the difference $(75 - 42)$ divisible by 3?
(A) Yes
(B) No
(C) Maybe
(D) Depends on the team's strategy
(E) Not enough information
- Q7.** A athlete won 99 medals. Is 99 divisible by 9?
(A) Yes
(B) No
(C) Maybe
(D) Depends on the medal type
(E) Not enough information
- Q8.** The number of spectators at a sports event is 240. Is 240 divisible by 10?
(A) No
(B) Yes
(C) Maybe
(D) Depends on the event type
(E) Not enough information

Open Ended Questions (FRQ)

- Q9.** A long-distance runner ran 237 kilometers. Use long division to find the quotient and remainder when 237 is divided by 12. Express the result as $12q + r$, where q is the quotient and r is the remainder.
- Q10.** In a 400-meter dash, a sprinter ran 312 meters in the first part of the race. If the remaining distance is divided into segments of 8 meters each, how many segments will the sprinter have to run to complete the race? Show your work using the division algorithm (*div* and *mod* operations).

Chef's Tips

- Tip 1: Encourage students to check their work using multiplication
- Tip 2: Remind students to apply the correct divisibility rules for each divisor
- Tip 3: Emphasize the importance of remainder in real-life applications